R-Programming

Experiment-1:

Addition:

```
num1=as.integer(readline(prompt = &quot;enter the first number:&quot;))
num2=as.integer(readline(prompt = &quot;enter the second number:&quot;))
num3=num1+num2
print(num3)
```

```
1:1  (Top Level)
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Enter first number: 4
Enter second number: 4
[1] 8
```

Result:

Thus the basic program addition is executed successfully.

Experiment-2:

Subraction:



Result:

Thus, the basic program subtraction is executed successfully.

Experiment-3:

Multiplication:



Result:

Thus, the basic program multiplication is executed successfully.

Experiment-4:

Division:

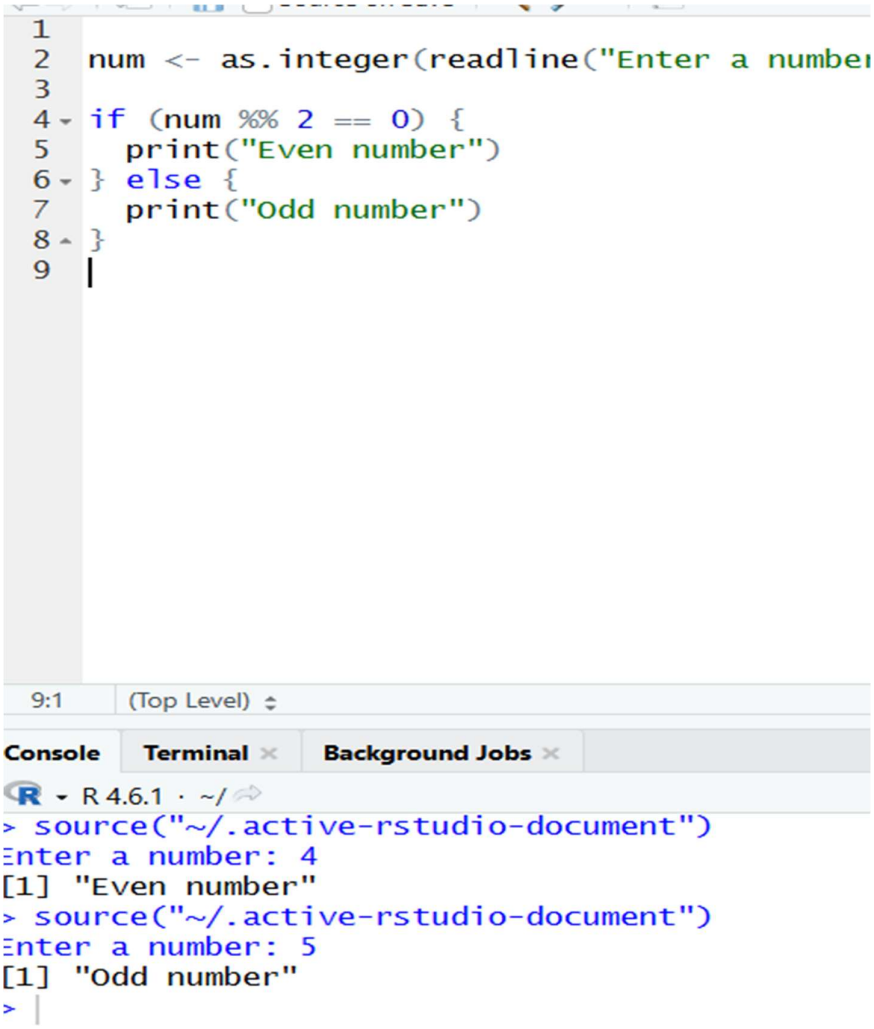


Result:

Thus, the basic program division is executed successfully.

Experiment-5:

Odd or Evevn:

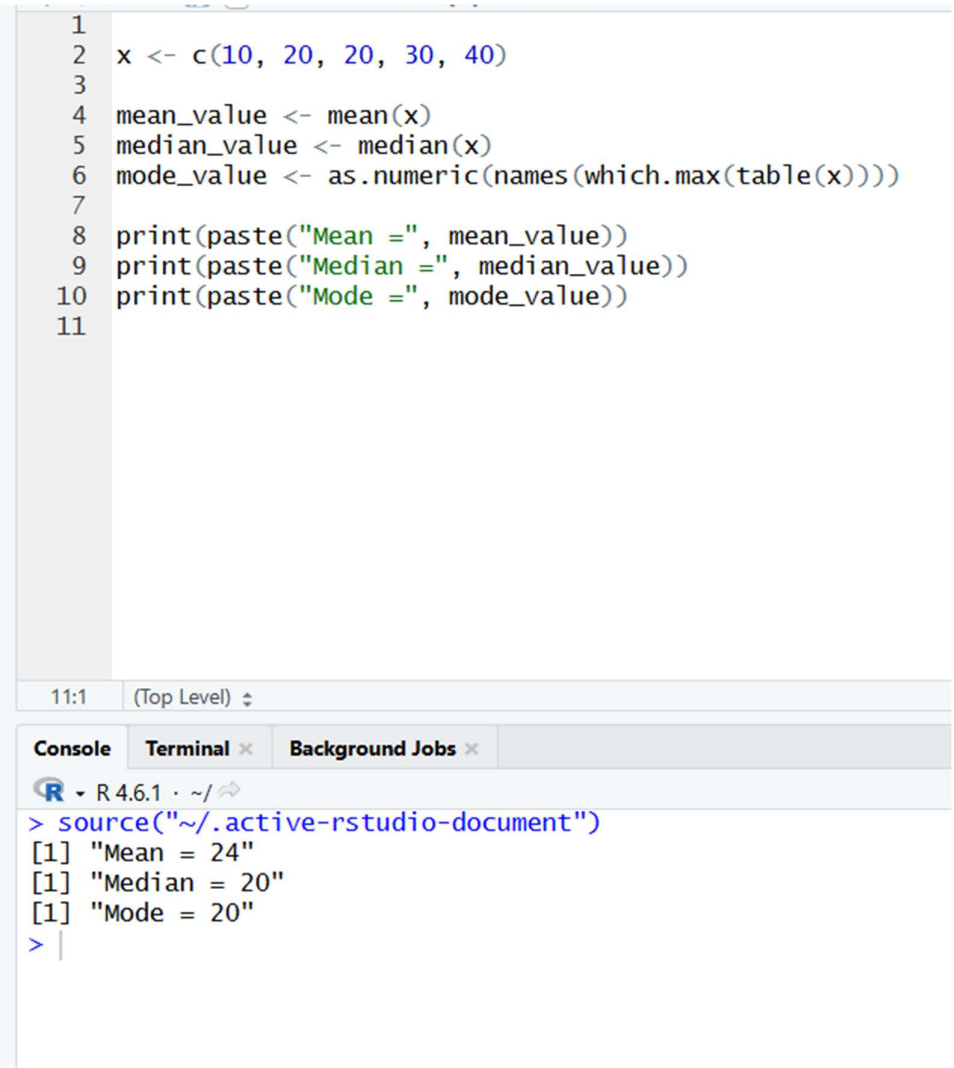


Result:

Thus, the basic program odd or even is executed successfully.

Experiment-6:

MEAN,MEDIAN,MODE:



Result:

Thus the central tendency and measure of dispersion is executed successfully.

Experiment-7:

Summary:

```
1
2  x <- c(10, 20, 20, 30, 40)
3
4  summary(x)
5
```

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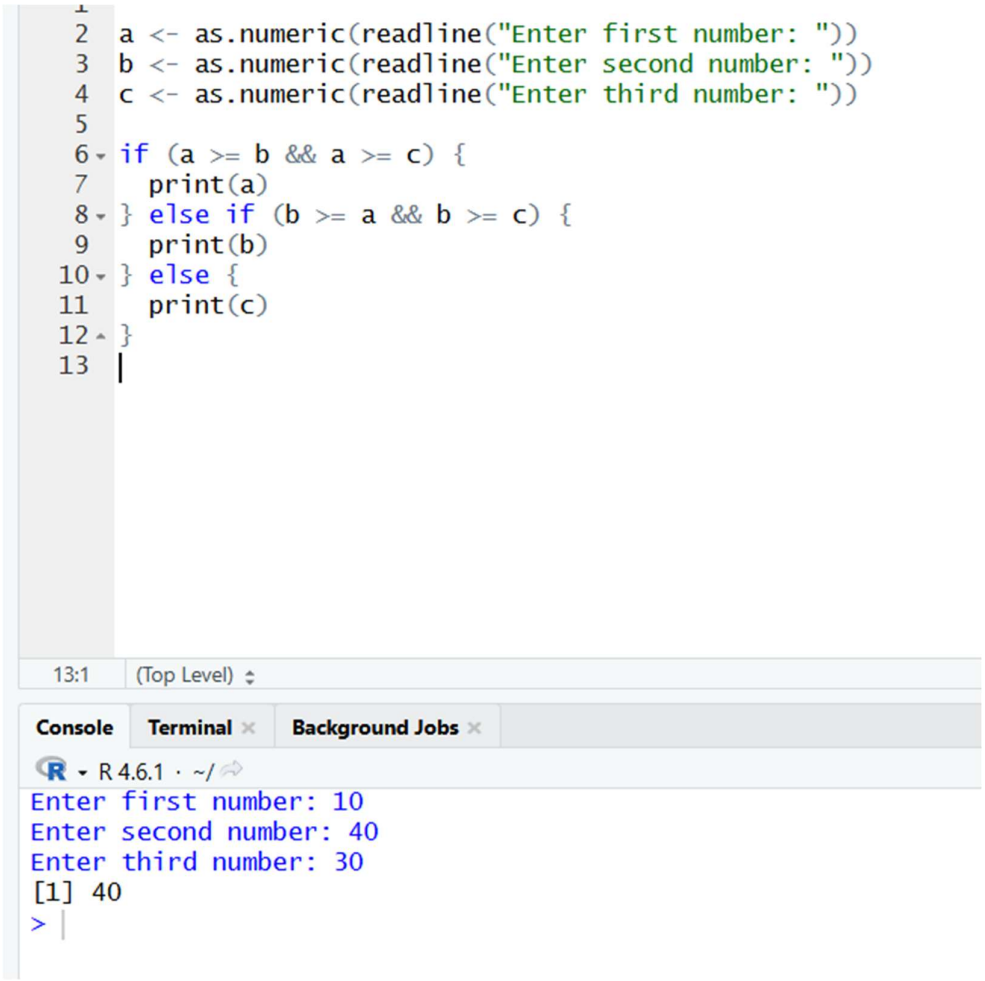
```
> source("~/.active-rstudio-document")
> x <- c(10, 20, 20, 30, 40)
>
> summary(x)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     10      20      20      24      30      40
>
```

Result:

Thus, the central tendancy and measure of dispersion is executed successfully.

Experiment-8:

GREATER AMONG THREE NUMBERS:



Result:

Thus, the greatest among the three numbers was executed successfully.

Experiment:9

IQR:

```
1
2  x <- c(10, 20, 20, 30, 40, 50, 60)
3
4  q1 <- quantile(x, 0.25)
5  q3 <- quantile(x, 0.75)
6  iqr_value <- IQR(x)
7
8  print(paste("Q1 =", q1))
9  print(paste("Q3 =", q3))
10 print(paste("IQR =", iqr_value))
11
```

10:33 (Top Level)

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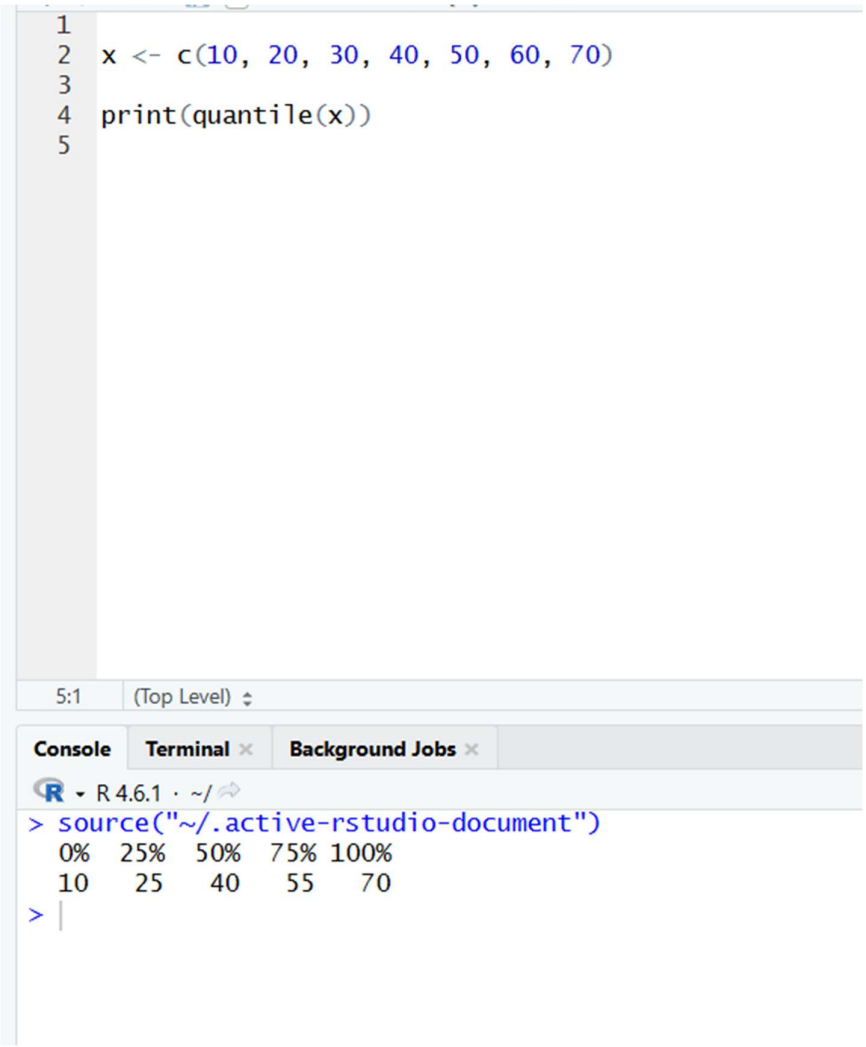
```
> source("~/.active-rstudio-document")
[1] "Q1 = 20"
[1] "Q3 = 45"
[1] "IQR = 25"
>
```

Result:

Thus, the program for central tendency and data dispersion measures was executed successfully.

Experiment-10:

QUANTILE:



Result:

Thus, the program for central tendency and data dispersion measures was executed successfully

Experiment-11:

MID RANGE:

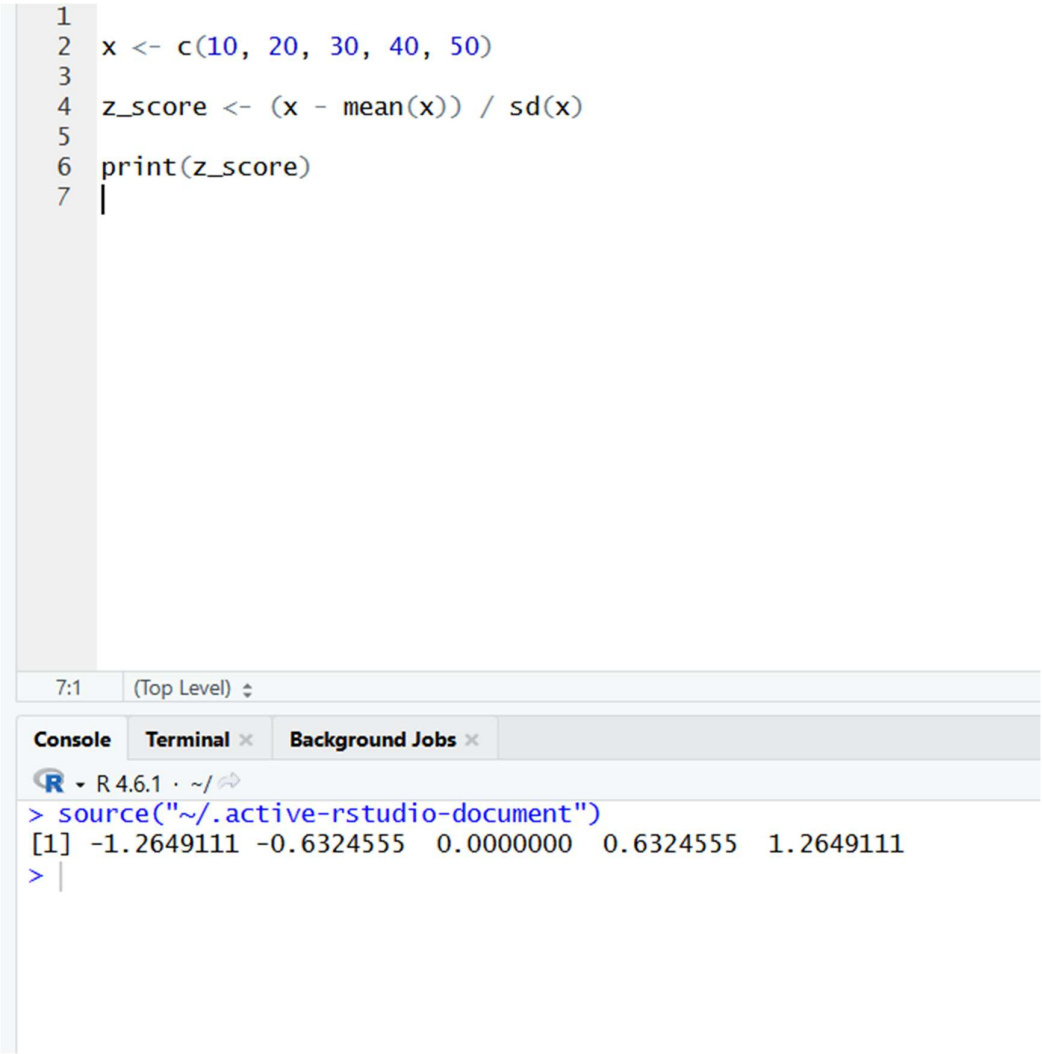


Result:

Thus, the program for central tendency and data dispersion measures was executed successfully.

Experiment-12:

Z-SCOORE NORMALIZATION:

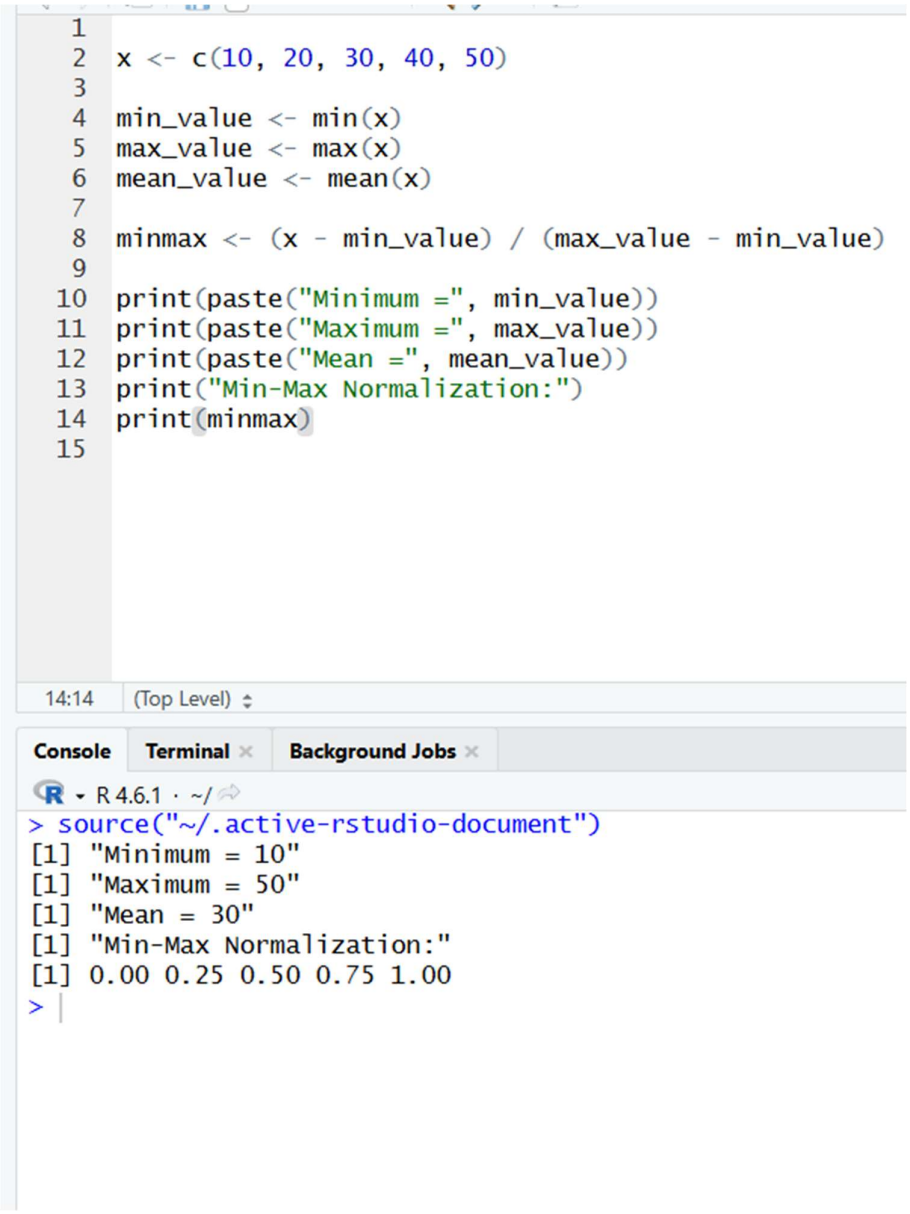


Result:

Thus, the Z-score normalization using R tool was executed successfully.

Experiment-13:

MIN,MAX,MEAN,MINMAX:



```r

x <- c(10, 20, 30, 40, 50)

min_value <- min(x)
max_value <- max(x)
mean_value <- mean(x)

minmax <- (x - min_value) / (max_value - min_value)

print(paste("Minimum =", min_value))
print(paste("Maximum =", max_value))
print(paste("Mean =", mean_value))
print("Min-Max Normalization:")
print(minmax)

```

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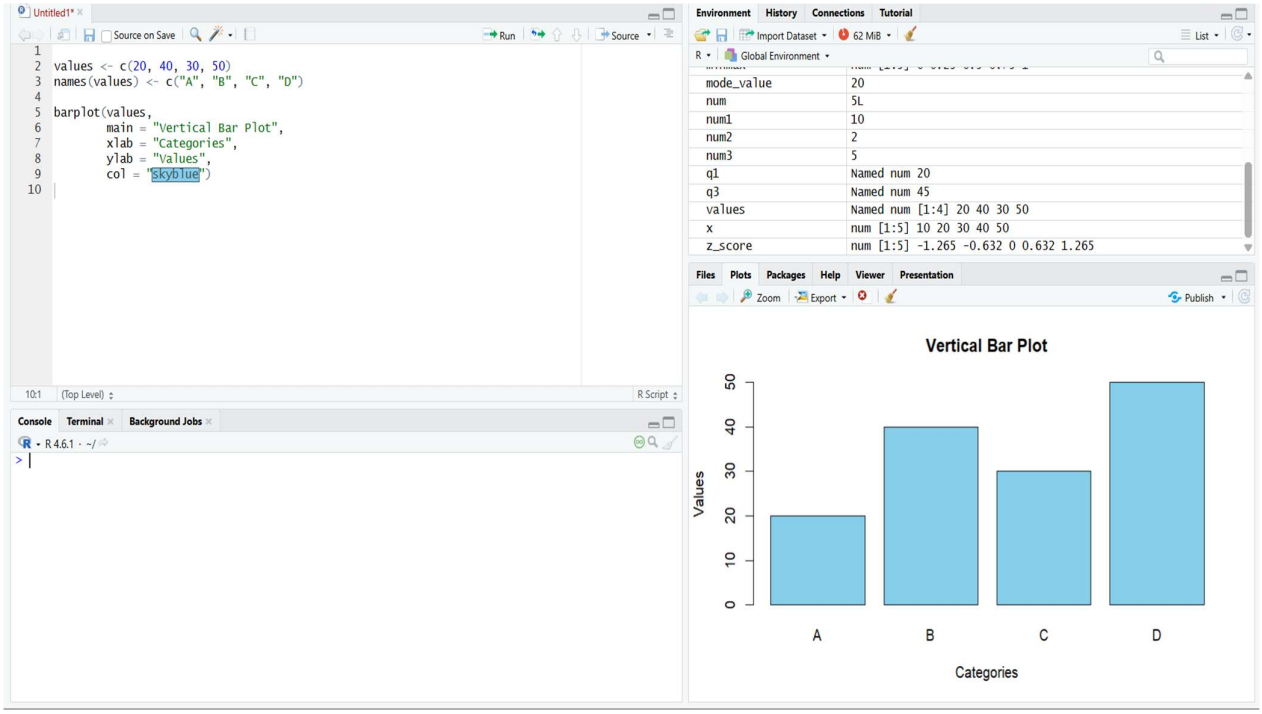
```
> source("~/.active-rstudio-document")
[1] "Minimum = 10"
[1] "Maximum = 50"
[1] "Mean = 30"
[1] "Min-Max Normalization:"
[1] 0.00 0.25 0.50 0.75 1.00
>
```

Result:

Thus, the program for min,max,minmax,mean was executed successfully.

Experiment-14:

BAR PLOT AND HORIZONTAL BAR:

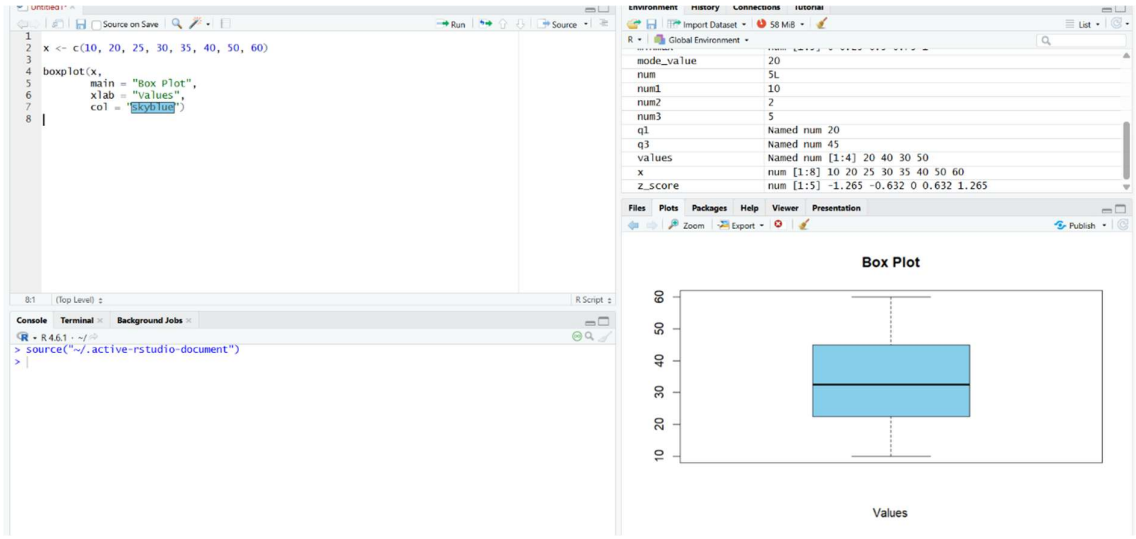


Result:

Thus, the bar and horizontal bar plot was executed successfully.

Experiment-15:

BOX PLOT:

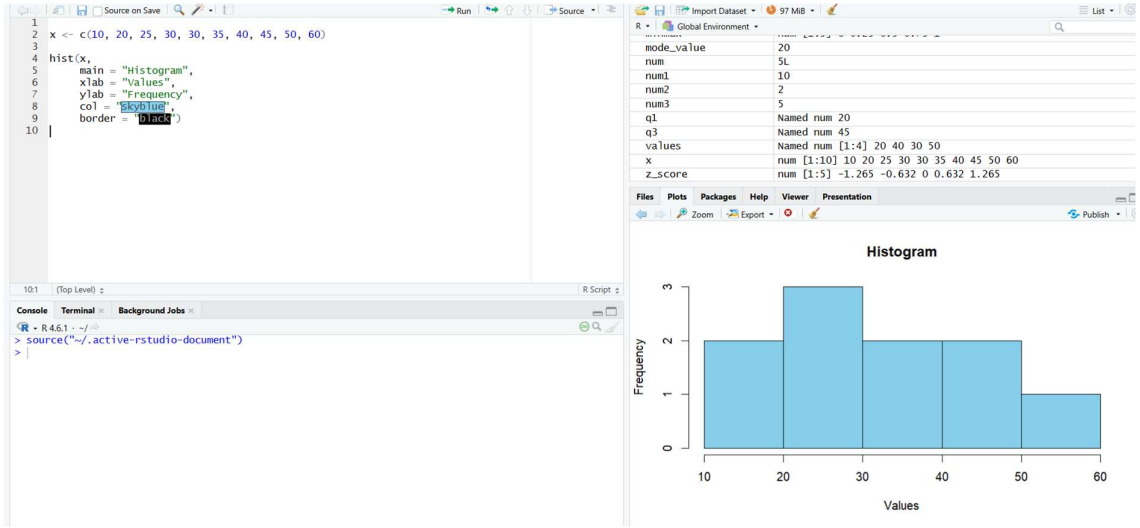


Result:

Thus, the box plot was executed successfully.

Experiment-16:

HISTOGRAM:

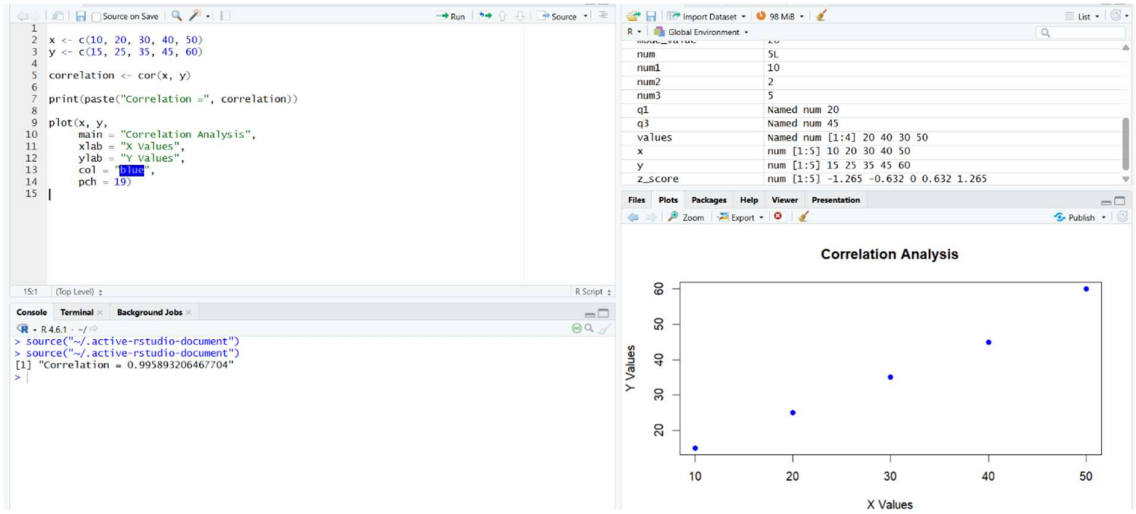


Result:

Thus, the histogram plot was executed successfully.

Experiment-17:

CORRELATION ANALYSIS:

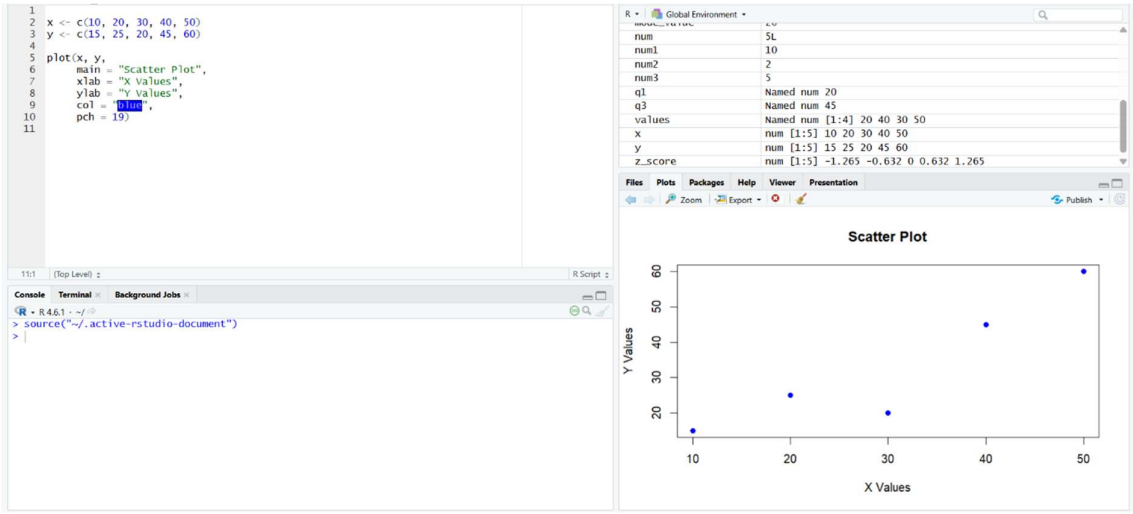


Result:

Thus, the correlation analysis was executed successfully.

Experiment-18:

SCATTER PLOT:

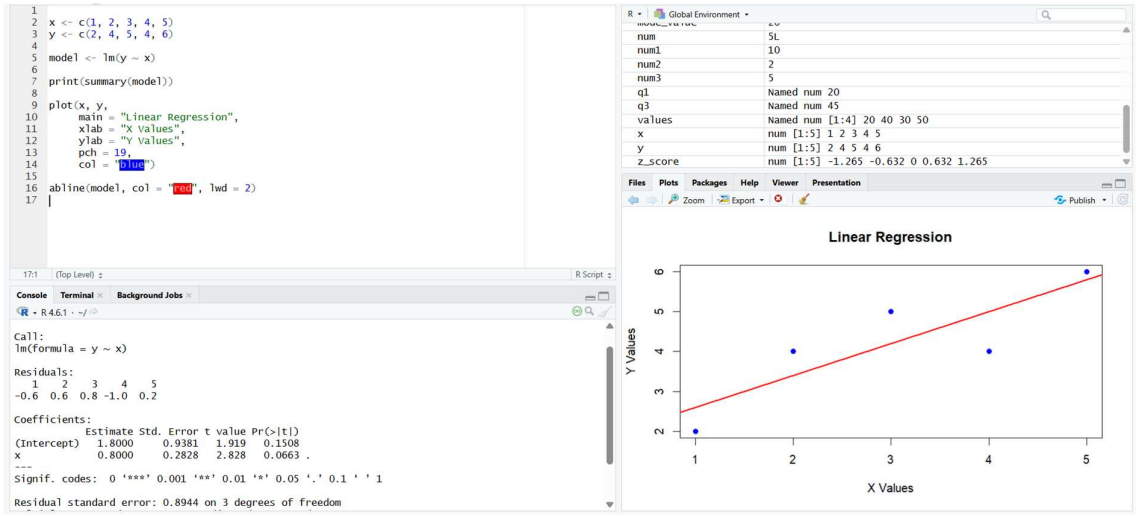


Result:

Thus, the scatter plot was executed successfully.

Experiment-19:

LINEAR REGRESSION:



Result:

Thus, the linear regression program was executed successfully.

Experiment-20:

MULTIPLE REGRESSION:

```
1
2  y <- c(10, 20, 30, 40, 50)
3  x1 <- c(1, 2, 3, 4, 5)
4  x2 <- c(2, 3, 5, 4, 6)
5
6  model <- lm(y ~ x1 + x2)
7
8  print(summary(model))
9
```

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```
Residuals:
         1          2          3          4          5
 1.030e-14 -1.269e-14 -1.040e-15 -1.040e-15  4.474e-15

Coefficients:
             Estimate Std. Error    t value Pr(>|t|)
(Intercept)  1.907e-14  1.697e-14  1.124e+00    0.378
x1           1.000e+01  8.727e-15  1.146e+15   <2e-16 ***
x2          -9.498e-16  8.727e-15 -1.090e-01    0.923
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.203e-14 on 2 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 3.456e+30 on 2 and 2 DF,  p-value: < 2.2e-16
```

Result:

Thus, the Multpile regression program was executed successfully.